

Photonic Optical Neural Network Inference Processor

Theoretical Foundations and a Latency Model for Light-Based Vector–Matrix Computation

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Abstract

A theoretical architecture is presented for performing the vector–matrix multiplication at the core of neural-network inference directly in the optical domain, using intensity superposition for addition and programmable attenuation for scalar multiplication. The underlying physical identities are derived from first principles; two illumination strategies (sequential and parallel) for realising a full matrix–vector product as a sum over time-integrated photocurrent are formalised; and it is shown that the rectification inherent to intensity photodetection provides a nonlinearity at no additional energetic cost. A multi-plane cascade architecture is then proposed in which successive layers’ weights are loaded onto physically distinct, static transmittance planes, removing the slow electro-optic settling time from the per-inference critical path entirely after a one-time loading step. A latency model is derived as a function of source modulation frequency and detector readout frequency, and evaluated against frequencies already demonstrated by commercially available spatial light modulators, yielding sub-millisecond projected inference times for a representative small embedded network. The principal limitation of the approach—*analog drift*—is discussed, and the architecture is argued to be best suited to small, embedded, low-power inference workloads such as those found in mobile robotics, rather than as a general-purpose accelerator. Finally, a prospective extension based on weight factorisation is outlined as an open direction for future exploration.

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1 Introduction

1.1 Motivation

The dominant cost of neural-network inference on conventional digital hardware is not arithmetic itself but the movement of data: weights and activations must be repeatedly fetched from memory, shuttled across buses, and accumulated in registers, each step dissipating energy as Joule heating in a resistive medium. This is sometimes referred to as the von Neumann bottleneck. For embedded and battery-powered platforms—small mobile robots, sensors, and other edge devices—the energy and thermal budget available for inference is often the binding constraint, more so than raw arithmetic throughput.

Optics offers a structurally different way of performing the same arithmetic. A beam of light carries information in its intensity and can be attenuated, combined, and detected with comparatively simple, passive, or near-passive components. Unlike electrical current, which dissipates energy proportionally to I^2R as it travels through a conductor, a free-space or guided optical signal can in principle traverse a static attenuating element and be summed at a detector with losses that do not scale with the *distance* the signal must travel, only with the optical elements it encounters. This motivates a longstanding line of research into *photonic* neural-network accelerators.

1.2 Related Work

Photonic implementations of neural-network primitives have been demonstrated across several physical substrates. Shen *et al.* realised matrix multiplication using cascaded Mach–Zehnder interferometers on a coherent nanophotonic circuit [1]. Tait *et al.* introduced the “broadcast-and-weight” protocol, in which a bank of tunable, real-valued (signed) optical weights multiplies a wavelength-multiplexed signal prior to detection on a single photodiode—a direct physical precedent for the signed-weight encoding discussed in Section 2 [3]. Lin *et al.* demonstrated an entirely passive, diffractive approach in which successive phase-modulating layers perform inference at the speed of light with no detector in the loop until the final output plane [2]. Hamerly *et al.* analysed the scaling of optical neural networks based on coherent photoelectric multiplication and argued for the energy advantages of optics at large fan-in [4]. More recently, integrated photonic tensor cores have demonstrated parallel convolutional processing on-chip [5], and wavelength-division multiplexed photonic accelerators have reported throughput in the range of several trillion operations per second [6]. The reader is referred to Shastri *et al.* for a broad review of photonic approaches to artificial intelligence and neuromorphic computing [7], and to Wright *et al.* for a treatment of physical neural networks—of which optical systems are one instance—trained directly through their own physical dynamics [8].

The present work does not introduce a new optical device or a new training algorithm. Its contribution is a self-contained, from-first-principles derivation of the arithmetic primitives (Section 2), an explicit comparison of two illumination strategies for realising the vector–matrix product with ordinary incoherent sources rather than coherent interferometry (Section 3), and a closed-form latency model that connects achievable inference speed directly to component frequencies that are commercially available today (Section 4), targeted specifically at the embedded, low-power regime rather than at data-centre-scale throughput.

1.3 Contributions

- A first-principles derivation of optical addition and optical scalar multiplication as the two arithmetic primitives required for inference (Section 2).

- A formalisation of signed-weight encoding via a differential (push–pull) channel pair (Section 2.3).
- Two illumination protocols—sequential and parallel—for realising a full layer’s vector–matrix product, with an explicit proof that the rectifying nonlinearity required between layers is a byproduct of intensity detection rather than an added component (Section 3).
- A proof that physically cascading multiple weight planes *without* an intervening detection-and-relaunch step collapses the network to a single linear transform, and a multi-plane pipeline architecture that avoids this while removing the dominant electro-optic settling latency from the per-inference critical path (Section 4).
- A closed-form latency model evaluated against frequencies already reported for commercially available spatial light modulators (Section 5).
- A prospective extension to weight factorisation as a path toward single-pixel-per-neuron encoding, with its principal limitations stated (Section 8).

2 Theoretical Foundations of Optical Arithmetic

A neuron computes $o = \sum_i x_i w_i$. Realising this physically requires exactly two operations: scalar multiplication of a signal by a weight, and summation of several such products. Both are derived here from the physics of incoherent light.

2.1 Optical Addition via Intensity Superposition

Consider two mutually incoherent optical sources of intensity I_1 and I_2 —mutual incoherence holds automatically for sources with independent phase noise, such as separate LEDs or laser diodes, and is the relevant regime for the architecture proposed here (as opposed to the coherent regime of [1], which requires interferometric phase stability). A photodetector does not respond to the electric field directly; its photocurrent is proportional to the time-averaged optical power incident on its active area,

$$i_{\text{photo}} = \mathcal{R} \int_A |E(\mathbf{r}, t)|^2 dA, \quad (1)$$

where $\mathcal{R}$ is the detector responsivity. For two incoherent fields, the cross term in $|E_1 + E_2|^2 = |E_1|^2 + |E_2|^2 + 2 \operatorname{Re}(E_1 E_2^*)$ time-averages to zero because the relative phase of E_1 and E_2 is random and uncorrelated with the integration window. The detector therefore sees

$$i_{\text{photo}} \propto I_1 + I_2. \quad (2)$$

This is the physical content of optical addition: *incoherent intensities add linearly at a square-law detector*. Concretely, a source of intensity $I_1 = 1$ and a second independent source of intensity $I_2 = 2$, both incident on the same detector, are read out as $I_1 + I_2 = 3$ —not because any computation has taken place, but because the detector physically cannot distinguish the origin of the power it absorbs. Equation (2) generalises immediately to N sources: $i_{\text{photo}} \propto \sum_{i=1}^N I_i$, with no dependence on N beyond the detector’s saturation limit.

2.2 Optical Scalar Multiplication via Attenuation

Let a beam of intensity I_{in} pass through a neutral attenuating element of power transmittance $T \in [0, 1]$. By definition of transmittance,

$$I_{\text{out}} = T \cdot I_{\text{in}}. \quad (3)$$

A source of intensity $I_{\text{in}} = 1$ passing through an element with $T = 0.5$ emerges with $I_{\text{out}} = 0.5$ —a physical scalar multiplication by a number in $[0, 1]$, realised with no active component. For a thin absorptive medium this transmittance follows the Beer–Lambert law, $T = e^{-\alpha \ell}$; for a programmable amplitude element such as a liquid-crystal cell or a digitally addressed micromirror operated in a time-averaged (pulse-width-modulated) regime, T is instead set directly by an applied control signal, and it is this latter regime that is assumed for the remainder of the paper, since it allows T to be set to an arbitrary value in $[0, 1]$ under digital control rather than being fixed by a material’s chemistry.

Combining (2) and (3), N independent sources $x_1, \dots, x_N$, each individually attenuated by a corresponding transmittance $w_1, \dots, w_N \in [0, 1]$ and incident on a common detector, produce a photocurrent proportional to

$$o = \sum_{i=1}^N x_i w_i, \quad (4)$$

which is precisely the multiply–accumulate (MAC) operation at the core of a neuron’s pre-activation. No electronic multiplication or addition has occurred; both operations are properties of how light propagates and how a photodetector integrates power.

2.3 Signed Weights via Differential Encoding

Equation (3) restricts w_i to $[0, 1]$: a passive attenuator can only remove energy, never add it, and so cannot directly represent a negative weight. This is resolved with a differential (push–pull) encoding, decomposing each signed weight as

$$w_i = w_i^+ - w_i^-, \quad w_i^+ = \max(w_i, 0), \quad w_i^- = \max(-w_i, 0), \quad w_i^+, w_i^- \in [0, 1]. \quad (5)$$

Each input x_i is routed through *two* attenuating elements in parallel, w_i^+ and w_i^- , and accumulated on two separate detectors D^+ and D^- following (4):

$$o^+ = \sum_i x_i w_i^+, \quad o^- = \sum_i x_i w_i^-. \quad (6)$$

The neuron’s true pre-activation is recovered by a single electronic (or balanced-photodetector) subtraction,

$$o = o^+ - o^- = \sum_i x_i (w_i^+ - w_i^-) = \sum_i x_i w_i. \quad (7)$$

This is the optical analogue of balanced detection—a longstanding technique in optical receivers for common-mode noise rejection—and is structurally similar to the signed weight banks of [3]. The cost is a doubling of the number of attenuating elements and detectors per layer; no other modification to the architecture of Section 3 is required.

3 Architecture: the Photonic Inference Processor

3.1 The Weight Plane as a Programmable Transmittance Matrix

The physical weight-storage element is idealised as a *transmittance plane*: a two-dimensional array of independently and continuously addressable cells, each cell (i, j) exposing a transmittance $T_{ij} = w_{ij} \in [0, 1]$ to any light incident upon it. For a layer mapping N inputs to M neurons, the plane is organised as an $N \times M$ grid, with column i corresponding to input x_i and row j corresponding to output neuron n_j , as in Table 1.

Table 1: Organisation of the weight plane for a layer with two inputs and three neurons. Column i holds every weight that input x_i is multiplied by; row j holds every weight contributing to neuron n_j .

	x_1	x_2
n_1	$W_{x_1 n_1}$	$W_{x_2 n_1}$
n_2	$W_{x_1 n_2}$	$W_{x_2 n_2}$
n_3	$W_{x_1 n_3}$	$W_{x_2 n_3}$

3.2 Vector–Matrix Multiplication by Column Illumination and Row Integration

Let column i of the plane be illuminated with an incident intensity x_i , attenuated cell-by-cell by the column’s transmittance values, so that light emerging from cell (i, j) carries intensity $x_i w_{ij}$ (Eq. (3)). Let row j of the plane be optically coupled to a single dedicated detector D_j —physically, a cylindrical (one-axis) imaging element whose focusing power lies along the column direction and whose null axis lies along the row direction, so that all light emerging from row j , regardless of which column it came from, is collected onto the same detector, while light from row $j' \neq j$ is kept spatially separated and collected onto $D_{j'}$. Under illumination of *all* N columns, detector D_j integrates, by Eq. (2),

$$o_j = \sum_{i=1}^N x_i w_{ij}, \quad (8)$$

which is exactly the pre-activation of neuron j . Two strategies exist for realising the required “illuminate all columns” condition.

3.2.1 Sequential (Time-Multiplexed) Illumination

A single source illuminates column i with intensity x_i for a short dwell time δt , then switches to column $i + 1$, and so on through $i = N$, while detector D_j integrates continuously across the entire sequence without being reset between columns. Because detection is a linear, time-integrating process (Section 2.1 holds equally for sources separated in time as for sources separated in space, since the detector’s integral is taken over the union of both), the accumulated photocurrent after the full sequence is exactly Eq. (8): the detector cannot distinguish “ N values arriving from different places at the same time” from “ N values arriving from the same place at different times.” This mode requires only a single source, at the cost of a total illumination time of $N \delta t$ that grows linearly with the number of inputs, and it requires the column-addressing mechanism (*e.g.* a beam deflector) to be synchronised with the source modulation.

3.2.2 Parallel (Broadcast) Illumination

Alternatively, N independent, fixed-position sources illuminate all N columns simultaneously and continuously, each source i permanently aligned with column i by construction rather than by active steering. Equation (8) is realised in a single exposure of duration equal to one source settling time, independent of N . This removes the $N \delta t$ scan term entirely and eliminates the need for any column-addressing synchronisation, at the cost of N physical sources rather than one. This mode is adopted as the preferred one for the latency model of Section 4, since the cost of additional fixed, low-power sources is small relative to the latency benefit, and because a larger number of statically aligned components is, in practice, more reproducible than a single component requiring active, precisely synchronised steering.

3.3 Nonlinearity at No Additional Cost

A photodetector measures power, and power is non-negative by construction: no physical detector can report $o_j < 0$. If the electronic readout circuit clamps its output at zero rather than reporting a calibration offset as negative, the measured quantity is exactly

$$\hat{o}_j = \max(o_j, 0) = \text{ReLU}(o_j). \quad (9)$$

The rectified linear unit, ubiquitous in modern neural networks, is therefore not an additional operation that must be engineered into the photonic pipeline—it is the default behaviour of an intensity detector, and the only design requirement is to *not* suppress it (for example, by avoiding any electronic offset that would allow a meaningful negative reading to survive past the detector stage). This is significant for Section 4: it identifies precisely which step in a multi-layer pipeline must remain electronic (the rectification, trivially) and which can in principle remain fully optical.

4 Cascade Architectures for Multi-Layer Inference

4.1 Why Naïve Optical Cascading Fails

It is tempting to chain L weight planes directly in the optical path—for instance with relay mirrors—so that a single input flash propagates through $\mathbf{W}_1$, then $\mathbf{W}_2$, $\dots$, then $\mathbf{W}_L$, and is read out only once at the end, never touching a detector in between. The following shows this discards the representational benefit of depth entirely.

Proposition 1. *Let $\mathbf{x} \in \mathbb{R}^{N_0}$ and let $f_l(\mathbf{h}) = \mathbf{h}\mathbf{W}_l$ for $l = 1, \dots, L$ denote L passive (no intervening nonlinearity) optical layers. Then $f_L \circ \dots \circ f_1(\mathbf{x}) = \mathbf{x}(\mathbf{W}_1\mathbf{W}_2 \cdots \mathbf{W}_L)$, i.e. the composition is itself a single linear map.*

Proof. By direct composition, $f_2(f_1(\mathbf{x})) = (\mathbf{x}\mathbf{W}_1)\mathbf{W}_2 = \mathbf{x}(\mathbf{W}_1\mathbf{W}_2)$, and by induction $f_L(\dots f_1(\mathbf{x})) = \mathbf{x}(\mathbf{W}_1\mathbf{W}_2 \cdots \mathbf{W}_L)$, which is linear in $\mathbf{x}$ with matrix $\mathbf{W}^* = \mathbf{W}_1\mathbf{W}_2 \cdots \mathbf{W}_L \in \mathbb{R}^{N_0 \times N_L}$. $\square$

Hence any L -plane cascade with no intervening detection step is, for every possible input, indistinguishable from a single plane programmed with $\mathbf{W}^*$. Depth contributes no additional expressive power; the nonlinearity of Eq. (9) is what makes depth meaningful, and it can only be applied at a detector. Any cascade architecture must therefore include a genuine optical-to-electronic-to-optical relaunch between planes that are meant to act as logically distinct layers, applying (9) at each relaunch.

4.2 The Multi-Plane Pipeline

The proposal is to assign each layer l its own physically distinct transmittance plane, loaded once with the (fixed, already-trained) weights $\mathbf{W}_l$ at deployment time, rather than reusing a single plane and rewriting it between layers and between successive inputs. The electro-optic settling time of a transmittance plane, $\mathcal{T}_{\text{settle}}$, is typically the slowest timescale in the system (Section 5); under the single-plane-reused scheme it is incurred L times *per inference*, since the plane must be erased and rewritten for every layer of every new input. Under the multi-plane scheme it is incurred L times *in total, once, at deployment*, and never again until the model is updated. Every subsequent inference pays only:

1. the parallel illumination time of plane l (Section 3.2.2), bounded by the source settling time $1/f_s$;
2. the readout time of the N_l detectors of plane l , bounded by the detector frequency f_d ;
3. the electronic application of Eq. (9) and the re-modulation of the source array driving plane $l + 1$, both of which are conventional electronic operations many orders of magnitude faster than $\mathcal{T}_{\text{settle}}$ and are neglected in the model below.

4.3 A Closed-Form Latency Model

For a network of L layers with N_l output neurons in layer l , operated in the multi-plane pipeline of Section 4.2 with parallel illumination (Section 3.2.2), the per-inference latency is

$$\mathcal{T}_{\text{total}}(f_s, f_d) = \sum_{l=1}^L \left(\frac{1}{f_s} + \frac{N_l}{f_d} \right) = \frac{L}{f_s} + \frac{1}{f_d} \sum_{l=1}^L N_l, \quad (10)$$

where f_s is the source array’s switching/settling frequency and f_d is the detector readout frequency, both assumed uniform across layers for simplicity. Under the single-plane-reused scheme, by contrast, the plane settling time appears once per layer per inference and dominates:

$$\mathcal{T}_{\text{total}}^{\text{reused}} = L \left(\mathcal{T}_{\text{settle}} + \frac{1}{f_s} + \frac{N_l}{f_d} \right) \approx L \mathcal{T}_{\text{settle}}, \quad (11)$$

since $\mathcal{T}_{\text{settle}}$ (milliseconds, for any technology with a mechanical or electro-optic relaxation step) is typically several orders of magnitude larger than $1/f_s$ or N_l/f_d (microseconds, for the electronic stages). Equation (10) versus (11) is the formal statement of why the multi-plane pipeline is the architecture of interest for repeated inference: it removes $\mathcal{T}_{\text{settle}}$ from the recurring critical path.

5 Numerical Projections

To ground Eq. (10) in currently available components, it is noted that digital micromirror devices (DMDs)—two-dimensional arrays of individually addressable, binary-state reflective elements, commercially available with over two million elements at resolutions up to full HD [9]—have been demonstrated switching binary patterns at 20 kHz, verified by direct photodiode measurement [10], with specific commercial controller chipsets rated up to 9 253 Hz for arbitrary binary patterns [9] and individual mirror transition times as short as 5–25 μs [11]; related work on dual-pass binary modulation has reported pattern rates up to 32 kHz [11], and high-pixel-count variants with reported refresh rates up to 50 kHz exist [12]. A value of $f_s = 20 \text{ kHz}$ is taken as a conservative,

already-demonstrated operating point for the source/plane switching stage. For the detector stage, silicon photodiodes with transimpedance front-ends routinely operate at $f_d = 1$ MHz or higher; this is adopted as a conservative value for f_d , well below the gigahertz bandwidths achievable with dedicated high-speed photoreceivers.

Figure 1 evaluates Eq. (10) over $f_s, f_d \in [10^3, 10^8]$ Hz for a representative small embedded network with layer widths 784-64-32-10 ($L = 3$, $\sum_l N_l = 106$), the input width chosen to match a 28×28 pixel image. At the conservative operating point identified above ($f_s = 20$ kHz, $f_d = 1$ MHz), Eq. (10) gives

$$\mathcal{T}_{\text{total}} = \frac{3}{20,000} + \frac{106}{1,000,000} \text{ s} = 256 \mu\text{s}, \quad (12)$$

i.e. an upper bound on throughput of approximately 3.9×10^3 inferences per second for this network, using only switching frequencies already reported for commercially available components, and without invoking the comparatively exotic and costly devices (high-power ultrafast lasers, cryogenic detectors) found in some data-centre-scale photonic accelerator proposals. This is presented as a latency *projection* from a simplified model, intended to establish plausibility and a useful order of magnitude, not a measured result; it neglects, in particular, settling transients in the source array and any analogue-to-digital conversion overhead in the rectification step.

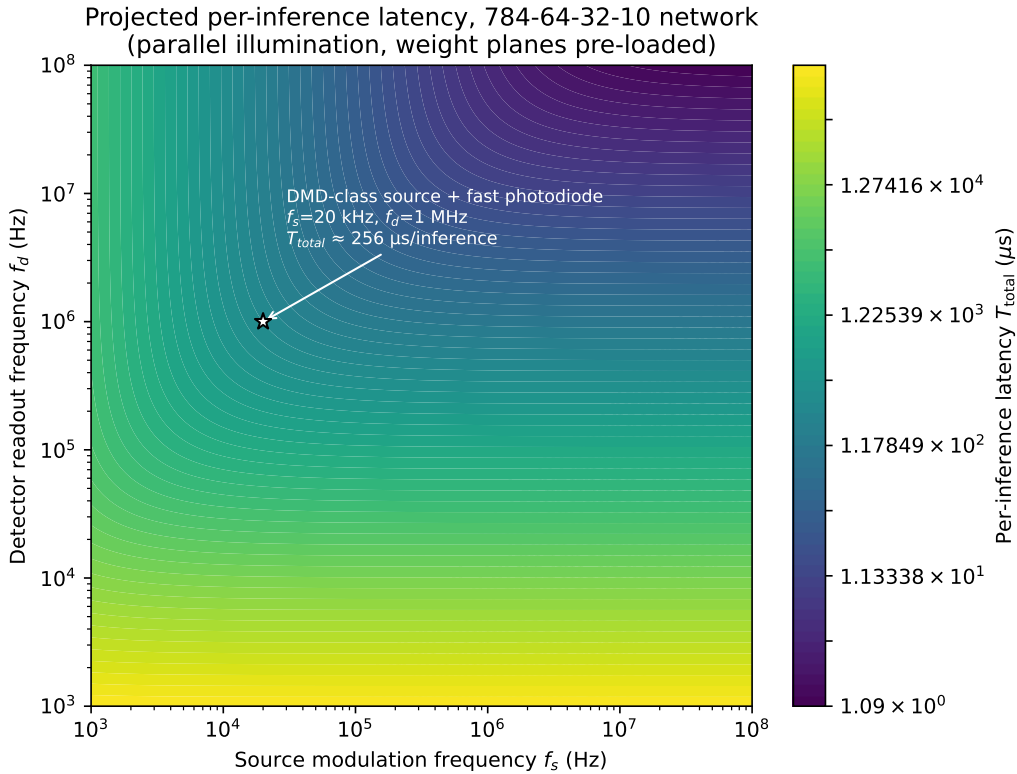


Figure 1: Projected per-inference latency of a 784-64-32-10 network operated in the multi-plane pipeline mode (Section 4.2), as a function of source switching frequency f_s and detector readout frequency f_d , from Eq. (10). The marked point corresponds to frequencies already demonstrated by commercially available digital micromirror devices and ordinary silicon photodiodes (Section 5).

The Python script used to generate Figure 1 from Eq. (10) is provided alongside this document

(`latency_heatmap.py`) so that the projection can be recomputed for other network shapes or component frequencies.

6 Practical Considerations and Scope

6.1 Analog Drift and Error Accumulation

Every stage of the architecture proposed here is analog: source intensity, plane transmittance, and detector responsivity are all continuous physical quantities subject to thermal drift, ageing, and manufacturing variation, none of which are corrected automatically as they would be by the discrete error-correcting margins of digital logic. Errors at each layer compound multiplicatively across L layers. This is a structural disadvantage relative to digital accelerators and is not addressed by any mechanism proposed in this paper; periodic recalibration against a known reference input is assumed necessary in any physical realisation.

This consideration directly motivates the target application domain. Small embedded models—few layers, narrow layers, the kind of network that might run on a microcontroller-class robotics platform rather than a data-centre accelerator—accumulate drift over fewer multiplicative stages and over a smaller weight count than large models, and are correspondingly more tolerant of a given per-weight error budget. The architecture is therefore positioned as a candidate for small, embedded, power-constrained inference rather than as a general substitute for large-scale digital accelerators, where the precision requirements of deep networks would demand a calibration overhead that this paper does not attempt to quantify.

6.2 Energy and Thermal Considerations

A central motivation for photonic approaches to neural computation is that optical signal combination (Section 2.1) and attenuation (Section 2.2) do not, in principle, dissipate energy in proportion to the distance a signal travels, unlike resistive electrical interconnect; the energy cost is instead dominated by the generation and detection of light at the endpoints of the computation. This motivates the broader interest in photonic accelerators surveyed in [7], and is one of the principal arguments for the scalability of optical neural networks at large fan-in in [4]. No quantitative energy claim is made for the specific architecture proposed here, since this would require a component-level power budget that is properly the subject of a hardware-implementation study rather than the present theoretical treatment; the direction of the argument is noted without asserting a specific efficiency figure.

6.3 Target Domain

Taken together, the latency projection of Section 5 (sub-millisecond inference for a small network, using already-available component speeds) and the drift considerations above point toward a specific niche: low-power, small-model, latency-tolerant-of-occasional-recalibration inference embedded directly on a robotic platform, where the relevant competitor is not a data-centre GPU but a low-power microcontroller, and where a modest, fixed accuracy budget (rather than the state-of-the-art accuracy of a large digital model) may be an acceptable trade for inference that consumes no dynamic switching energy in its arithmetic core.

7 Limitations and Open Problems

- **Weight resolution.** The number of distinguishable transmittance levels a given plane technology can realise, and how that number degrades under the detector noise floor, bounds the effective bit-width of every weight; this paper does not derive that bound for any specific plane technology.
- **Calibration.** The differential encoding of Section 2.3 and the rectification of Section 3.3 both assume a stable, known zero-point; drift in that zero-point is indistinguishable from drift in the weights themselves without a separate reference channel.
- **Fan-out.** Section 3.2.2 assumes N sources can each be made to illuminate their column with reasonably uniform intensity across all M rows; the optics required to do so without crosstalk to neighbouring columns has not been analysed here.
- **Training.** This paper assumes weights are trained conventionally, off-line, and only the resulting fixed matrix is loaded onto the device; in-situ, hardware-aware training in the presence of the non-idealities above is left to future work.

8 Further Perspectives: Weight Factorisation and Single-Pixel-per-Neuron Encoding

The architecture of Section 3 requires one transmittance cell per (input, neuron) pair: a layer mapping N inputs to M neurons occupies an $N \times M$ region of the plane. For large layers—in particular the first layer of a network processing raw pixel images ($N = 784$ for MNIST, $N = 3072$ for CIFAR-10)—this places non-trivial demands on the physical extent of the transmittance plane. An algebraic observation suggests a qualitatively different encoding that would, in principle, reduce that demand to *one pixel per neuron*, though at a cost discussed below.

8.1 The Factorisation Idea

Consider a single neuron j computing $z_j = \sum_{i=1}^N x_i w_{ij}$. Suppose the weights can be written (exactly, or approximately) as

$$w_{ij} \approx \alpha_j \cdot k_{ij}, \quad k_{ij} \in \mathbb{Z}_{\geq 0}, \quad (13)$$

for some scalar $\alpha_j \in [0, 1]$ and non-negative integers k_{ij} . Then

$$z_j \approx \alpha_j \sum_{i=1}^N k_{ij} x_i. \quad (14)$$

Optically, this is implemented as follows: the transmittance plane assigns neuron j a *single cell* with transmittance α_j , encoding the common scale factor. The integer ratio k_{ij} is encoded not in the plane but in time: source i emits exactly k_{ij} successive flashes of intensity x_i into the cell while the detector integrates without reset. After all $N \cdot K$ flashes (where $K = \max_i k_{ij}$), the detector’s accumulated photocurrent is proportional to $\alpha_j \sum_i k_{ij} x_i = z_j$, recovering the neuron’s pre-activation from Eq. (14).

In this encoding a full HD transmittance plane (1920×1080) can, in principle, host up to $\approx 2,073,600$ neurons simultaneously, one per physical pixel, regardless of the fan-in N —since the fan-in is absorbed into the temporal flash sequence rather than into the spatial extent of the plane.

8.2 Principal Limitations

The approach is mathematically sound and is structurally related to the per-channel scale factors used in quantised neural networks (where weights are stored as integers multiplied by a per-layer or per-channel floating-point scale). However, two fundamental obstacles arise when one attempts to exploit it within the parallel optical framework of Section 3.

Loss of inter-neuron parallelism. In the baseline architecture, all M neurons of a layer are computed simultaneously because every row’s detector integrates independently. In the factorised encoding, neuron j requires k_{ij} flashes of source i , while neuron j' may require $k_{ij'} \neq k_{ij}$ flashes of the same source. If both neurons are processed at the same time, the single source i cannot simultaneously deliver k_{ij} flashes to cell j and $k_{ij'}$ flashes to cell j' : once the flash reaches the transmittance plane, it illuminates both cells simultaneously, with no mechanism to stop one cell from accumulating while the other continues. Restoring selectivity would require per-pixel electronic shuttering—reintroducing exactly the element-level addressing complexity that the optical approach was designed to avoid. Without it, neurons can only be computed one at a time, eliminating the parallelism that constitutes the primary advantage of the optical architecture over a software loop.

Precision penalty. Approximating real-valued weights w_{ij} to the form $\alpha_j k_{ij}$ with integer k_{ij} requires rounding onto a grid. With b -bit quantisation, the average value of k_{ij} is of order 2^{b-1} . For a neuron with N inputs, the total number of flashes required to compute a single pre-activation is therefore $\sum_i k_{ij} \approx N \cdot 2^{b-1}$. For a first layer of a network processing MNIST images ($N = 784$) with 8-bit weight resolution ($b = 8$), this amounts to approximately $784 \times 128 \approx 100,000$ flashes per neuron—compared to N source activations (one per input) in the parallel baseline. The temporal overhead scales as $N \cdot 2^{b-1}$ rather than the baseline’s $O(1)$, which reverses the latency advantage for any practically useful level of precision.

8.3 Status

The factorisation idea is recorded here as an open direction rather than a solved problem. Its merit is that it reduces the *spatial* demand on the transmittance plane from $O(N \times M)$ cells to $O(M)$ cells (one per neuron), which is a genuine structural advantage for very wide input layers. Whether the temporal overhead can be brought within acceptable bounds—for example by exploiting sparsity in the weight matrix, using very low-bit quantisation, or combining factorisation with partial spatial parallelism—remains to be determined.

9 Conclusion

The foregoing has derived, from the physics of incoherent light detection, the two arithmetic primitives—addition by superposition and scalar multiplication by attenuation—sufficient to realise the multiply-accumulate operation of a neuron; shown that the rectifying nonlinearity required between layers is an intrinsic property of intensity detection rather than an added cost; and proved that a multi-layer cascade must include a genuine detection step between layers or else collapse to a single linear map. Building on these foundations, a multi-plane pipeline architecture is proposed in which the slow electro-optic settling time of the weight planes is incurred once, at deployment, rather than on every inference, and a closed-form latency model is derived and evaluated against frequencies already demonstrated by commercially available spatial light modulators and ordinary photodiodes, projecting sub-millisecond inference for a small embedded network. The principal open problem is analog drift, which makes the architecture best suited to small, embedded, low-power inference rather than to large-scale, high-precision workloads. A prospective weight-factorisation

scheme that would reduce the spatial footprint of the transmittance plane to one pixel per neuron has been outlined, along with the parallelism and precision trade-offs that must be resolved before it can be considered a practical alternative.

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